

Learning to choose, then navigate

Experiment 2 asks whether an explicitly trained controller can associate a visual pattern with reward and learn routes to food in an editable maze. A two-dimensional fly avatar makes the decisions visible. This is an engineering experiment that uses a fixed connectome-derived sensory encoder; it is not a claim that biological fly memories have been reconstructed.

Prospective status

Written on 20 September 2026 before implementation of this learning mechanism and before its training outcomes were observed. Numerical verification and exploratory development follow. This is a dated local protocol, not an external preregistration or a guarantee of success.

Preserve Experiment 1

Experiment 1 keeps its current launcher, UI, dynamics, graph and PDFs. The local Git tag exp-1-v0.2 points to commit 53dff10. Experiment 2 has setup_exp_2.sh, its own service and port 8767, separate run records and an independent interface. The existing Mac PLAN folder gains only this exp_2.pdf.

The controller learns. Connectome edges and neural model parameters remain fixed. A successful food-finding route alone does not demonstrate biological learning, cognitive understanding or an advantage from the fly wiring.

The first maze task

The arena is a 9 by 9 grid. Walls block movement. The user can edit walls, the starting cell and two food sites. The agent has four actions: north, east, south and west. One action moves one grid cell. The rendered body is a schematic avatar, not a biomechanical fly.

The two sites display Pattern A (vertical bars) and Pattern B (horizontal bars). Their assignment swaps across episodes using a reproducible, balanced schedule. The rewarded pattern is configured before the experiment; the controller is not given the reward label. Contact with either food site ends the episode.

A correct food contact yields +1; the other pattern yields -0.25. Each ordinary movement costs -0.01. An attempted manual move into a wall costs -0.04. Autonomous policies see adjacent walls and choose among legal moves. The episode times out after 120 actions.

What the agent observes

Its engineered grid position, legal adjacent movements, and both food-site pattern signals are available throughout an episode. The visual beacons are readable across the arena and are not occluded by walls. The controller receives decoded sensory tokens; it receives no shortest path, food payoff label or distance-to-goal oracle.

This first task is route learning with visible beacons and perfect grid localization. It does not yet model an unknown maze, realistic fly vision, smell, fog of war or a general search strategy. Transfer tests explicitly check the limits of route memory.

From patterns to neural signals

In real-data mode, load the same 166,700-node MaleCNS-derived graph and fixed LIF equations used in Experiment 1. Feed a 32 by 24 pattern into mapped L1 columns on both sides, using the existing exploratory coordinate mapping and a drive of 0.6 mV per tick. Each pattern probe starts from reset and runs for 40 ms.

Pool measured L1 spike rates into a 4 by 4 grid for each eye: 32 neural features. Preserve the feature vectors, whole-network counts, voltage summaries, mapping identity, model parameters and graph checksums. Demonstrate distinguishable, finite responses before allowing neural-cue training.

The two known pattern responses become fixed calibration templates. Classification chooses the nearest response template. This is an engineered decoder calibrated on the two stimuli; it is not a discovered natural motor or recognition circuit.

Exact cached probes, not a continuous brain

Because this task repeats only two deterministic images, compute each reset response once and reuse its recorded features by image identity. The UI labels the neural plot as a cached probe. No new full-brain simulation occurs on every movement, and no continuous neural state connects maze steps.

On a Mac the portable default uses the explicitly synthetic teaching graph and CPU solver. On gx10-b the actual MaleCNS probes run on CUDA. The small Q-learning controller and maze run on the same server CPU. Never display cached traces as current whole-brain activity.

The learning rule

Use tabular Q-learning. $Q(s,a)$ estimates the discounted future return of action a in state s . Here s contains the grid position and the decoded pattern at each site. The Q table has four sensory combinations, 81 positions and four actions. Walls and the reward rule are fixed within a training experiment.

One-step update

$Q(s,a)$ becomes $Q(s,a) + \alpha * [r + \gamma * \max Q(\text{next_s}, \text{legal_action}) - Q(s,a)]$. For a terminal food contact or timeout, use r alone as the target. The learned quantities are entries in this external table; no synapse in the connectome is updated. [1]

Defaults: $\alpha = 0.25$; $\gamma = 0.95$. Exploration probability ϵ is $\max(0.05, 0.8 * 0.99^{\text{episode}})$. During training, choose a uniformly random legal action with probability ϵ ; otherwise choose a maximum- Q legal action, breaking ties reproducibly at random. Evaluation uses ϵ zero and never updates Q .

The step cost encourages shorter successful routes without revealing their direction. Exploration can initially look inefficient. A finite run with a fixed learning rate does not inherit an unconditional mathematical guarantee of optimality.

Resetting a trial preserves Q . Starting a new experiment or applying edited geometry starts fresh Q tables and a new run record. This prevents mixing incompatible reward tasks in one learning curve.

Controls that matter

The interface shows an editable maze, a six-legged fly avatar, its trajectory, visited cells and food patterns. The side panels show current reward, episode, exploration, action values, neural template features and the learning curve.

Train, Pause and Step

Train adds a chosen number of episodes to a bounded training queue. Speed controls display pacing in actions per second, not a biological time constant or a learning parameter. Pause keeps Q and the current trial. Step makes one learner decision and applies its update.

Manual movement

The arrow controls move the avatar while paused. Manual trials do not update the trained Q table or enter the autonomous training-success metric. Reset trial starts a clean autonomous trial while retaining learned Q. Finish/save preserves the experiment checkpoint.

Edit the arena

Select Wall, Erase, Start, Food site 1 or Food site 2, then click cells. Draft edits take effect only when Apply new experiment is pressed. Validation requires a closed boundary, distinct open markers and routes from the start to both sites without needing to pass through the other terminal site.

New experiment and reward reversal

New maze drafts a different layout. The rewarded-pattern selector and seed apply when creating a new experiment. Changing the reward identity resets learning; studying reversal while retaining memory requires a separate future protocol amendment.

Controlled evaluation

Every training episode has a matching image-cue control episode. That controller sees the same grid position and legal movements and uses the same Q-learning rule, initial values, exploration schedule, reward scheme and episode random seed. It recognizes the two rendered patterns directly, bypassing the neural simulation.

The image-cue control receives equivalent pattern information. If its decoded tokens match the neural templates, equal performance is expected. That outcome means the maze task and Q-learning worked, without evidence that the connectome contributed an advantage.

Evaluate freezes a policy checkpoint and runs all three agents: trained neural-cue Q, trained image-cue Q, and uniformly random legal movement. Use a disjoint seed namespace for evaluation. Report the trained maze separately from newly generated mazes with the same grid size and marker conventions.

Default evaluation uses 30 episodes per environment group and per agent. Repeat food-pattern swaps in balanced pairs. No weight changes occur during either familiar-maze or unseen-maze evaluation. Evaluation results cannot be appended to the training curve as if they were training trials.

Unseen mazes may expose strong overfitting because Q is indexed by coordinates. Failure to transfer is a valid result. This first protocol cannot establish a universal or best maze-scanning strategy.

What to measure

Primary outcomes: fraction of episodes reaching the rewarded pattern and mean return across all episodes. Also report steps, food mistakes and timeouts. Show a rolling training success rate using the last 20 completed autonomous episodes; label its actual sample count.

During evaluation, preserve per-trial outcomes and report successes divided by n with a Wilson 95 percent interval. This interval describes the finite trial sample, not variation across training seeds or hardware. A single training seed is an exploratory demonstration.

For successful episodes, shortest-path efficiency is shortest allowed start-to-goal length divided by steps taken. The shortest-path calculation belongs only to the evaluator; the policy never receives it. Paths through the wrong terminal food cell are excluded.

For a stronger follow-up, use at least five independently trained seeds, summarize per-seed results, freeze hyperparameters before evaluating and test on additional held-out layouts. Do not claim significance from hundreds of steps of the same policy as if each were an independent training experiment.

Development acceptance

The system must demonstrate actual reward-driven Q updates, checkpoint restoration, unchanged Q during evaluation, correctly excluded manual trials and preserved Experiment 1. Record favorable and unfavorable results. Performance comparisons must name the dataset, trained component and control.

Installation and two-machine use

From a cloned project, run `sh setup_exp_2.sh`. This creates or reuses the isolated runtime, prepares the labeled synthetic graph and starts Experiment 2 on `http://127.0.0.1:8767/`. Use `--no-start` to install and verify without launching. It does not start or modify Experiment 1.

For real data on a CPU, use `sh setup_exp_2.sh --dataset male-cns --backend cpu`. For a compatible Linux NVIDIA system, use `--dataset male-cns --backend cuda`. Python 3.11-3.13 and the same pinned dependencies apply. The full-data memory and download requirements remain those in the handbook.

In this lab, keep Experiment 1 on `gx10-a` at port 8765. Run Experiment 2 on `gx10-b` at port 8767 and forward only loopback to the Mac. The machines run separate experiments; no distributed simulation is claimed. Both receive matching source releases.

The launcher accepts `--port`, `--venv`, `--data-dir` and `--runs-dir`. Its default records directory is `runs/exp_2`. A busy port produces an actionable error; existing processes are not terminated. Cached neural probes are generated from the selected graph, never substituted from another dataset.

The user-facing maze is rendered in the Mac browser. Numerical probes and all experiment state are owned by the selected server. Refreshing the browser does not reset learning. Two tabs of one service share the same experiment.

Records, checkpoints and restoration

Each experiment gets a unique directory containing `source_manifest.json`, `manifest.json`, `config.json`, `commands.jsonl`, `episodes.jsonl`, `evaluations.json` and `checkpoint.npz`. Neural probe vectors and summaries are copied into the record with graph and mapping hashes; source code is preserved once per source identity.

The checkpoint stores both learned Q tables, completed episode count, validated layout/configuration and encoder identity. It is saved after each completed episode and on pause/save. Loading verifies dataset/encoder identity and opens a new run linked to the parent checkpoint, preserving the original record.

Checkpoint restoration resumes from a fresh episode at the boundary. It does not restore a partially completed maze trajectory or a continuously running brain, because those are not the state represented by this checkpoint. The saved episode count preserves the exploration schedule.

Each evaluation records its policy digest, seeds, layouts and trial outcomes. Check Q arrays before and after evaluation to reject accidental training leakage. Manual interventions are marked separately. Partial or failed runs retain their evidence and an explicit status.

Reproduction

Preserve source archives, graph data/checksums, neural probe artifacts, seed/configuration and policy checkpoints. A replay of the external controller can reuse verified probe features; reproducing the neuroscience probe additionally needs the graph and numerical backend. Dataset downloads and run records remain outside Git.

Interpretation and next stages

A higher reward rate on the training maze shows a controller adapted to this finite task. It does not show that the MaleCNS organism learned, that fly synapses changed, or that the avatar has natural fly behavior. The interface must keep the trainable controller and fixed neural encoder visibly separate.

A similar result for the image-cue baseline is informative. It identifies the ordinary learner as sufficient for the task. A claimed benefit from fly anatomy would need harder matched tests and appropriate wiring perturbations, with encoder information and training budgets controlled.

After this version: (1) test unseen-layout transfer, (2) add partial visibility and explicit agent memory, (3) consider an olfactory search task, or (4) introduce a stated mushroom-body-inspired plasticity rule. Each addition changes the scientific question and requires a dated amendment before evaluating its claims.

Published embodied fly platforms provide useful examples of sensory interfaces and trained controllers. They do not make our particular graph, dynamics or Q table biologically validated. Connectome-constrained visual work also fits physiological parameters; we are not adopting those fitted dynamics automatically. [2,3]

Success is measured improvement with reproducible records and appropriate controls. If the best observed strategy memorizes this maze, report memorization. If transfer or biological specificity is absent, report that explicitly.

References and protocol boundary

[01] Watkins and Dayan. Q-learning. Machine Learning (1992).

<https://www.gatsby.ucl.ac.uk/~dayan/papers/cjch.pdf>

[02] Wang-Chen et al. NeuroMechFly v2: simulating embodied sensorimotor control in adult Drosophila. Nature Methods (2024).

<https://www.nature.com/articles/s41592-024-02497-y>

[03] Lappalainen et al. Connectome-constrained networks predict neural activity across the fly visual system. Nature (2024).

<https://www.nature.com/articles/s41586-024-07939-3>

[04] MaleCNS official download inventory.

<https://male-cns.janelia.org/download/>

The official MaleCNS release supplies anatomical data. The current project supplies assumed dynamics, direct L1 stimulation, calibrated template readout, grid localization, reward values and the external learning mechanism. These layers must remain distinguishable.

Protocol v0.1 is the pre-implementation specification. Implementation checks and observed development results belong in STATUS.md and run records. A later document revision must identify what was observed before it was written. Keep one current exp_2.pdf in PLAN and preserve prior source versions in Git.

Amendment: a fair comparison

21 September 2026. This amendment is written before implementing and observing the new trainable-circuit benchmark. The original 20 September maze runs were already observed: fixed-neural-cue and image-cue Q tables were identical, with 30/30 familiar successes and 8/30 on unseen layouts in one development run. That is evidence of no added value from the cached visual probes in that implementation.

Two different questions

Part A audits the existing Exp 2 mechanism across ten seeds and both rewarded patterns: does the measured neural encoder change any decoded observation, Q update or outcome compared with direct image cues? Both use identical episode assignments, random decisions, rewards, hyperparameters and training budgets.

Part B tests whether a reduced network constrained by actual fly connections can learn action values better than a Q table under identical collected experience. It does not train the complete 166,700-neuron LIF model. A separate benchmark preserves the existing UI, sessions and graph files.

Q-learning is a learning algorithm; a connectome is a connectivity structure. Our neural controllers also learn action values using temporal-difference learning. The comparison is tabular versus neural representation, and anatomical versus randomized connectivity - not biology versus all reinforcement learning.

Exactly what will be trained

Load the verified MaleCNS graph and its annotations. Identify Kenyon_Cell and MBON classes. Select 256 connected Kenyon cells using a fixed anatomy-only random seed, retain their recorded positive-weight connections to MBONs, and discard MBON rows without a selected connection. Save every body ID, edge mask and initial weight. No behavioral outcomes influence circuit selection.

This deliberately reduced feedforward rate model excludes recurrent, modulatory and other brain pathways. Recorded contact counts initialize positive strengths, normalized to unit incoming sum. The imported positive sign is a modeling assumption. It is not the unchanged full-brain spiking simulation and is not validated mushroom-body physiology.

All agents observe grid x/y, the same two pattern identities and four legal-move bits. No rewarded-pattern label, complete maze, goal label, distance-to-food or shortest path enters the observation. A fixed random projection encodes one-hot x/y and both pattern labels plus legal moves into 256 nonnegative Kenyon-like activities. This projection is engineered, not recorded fly sensory wiring.

MBON-like activity = $\tanh(W \text{ times KC activity})$. Four learned linear readouts produce north/east/south/west action values. In the plastic variant, reward-driven TD errors update existing entries of W and the readout; absent edges stay zero and strengths stay nonnegative. In the frozen-circuit variant, only the readout changes.

Updating modeled KC-to-MBON strengths is genuine parameter training in this reduced artificial network. It is not proof of biological synaptic plasticity. Backpropagation/Adam and the external action readout are engineered learning mechanisms.
[5]

Controls and equal experience

Controllers in Part B: (1) tabular Q-learning, (2) recorded circuit with a trainable readout and frozen internal weights, (3) recorded circuit with trainable internal weights and readout, (4) randomized circuit with the same training mechanism and number of parameters. A random-action policy is an evaluation reference.

The randomized control uses directed bipartite edge swaps preserving incoming and outgoing degrees. It retains the same selected cells, input projection, readout initialization, positive weight multiset per output cell and edge count. The source and randomized masks and their difference are recorded. This isolates topology more closely than an unmatched dense ANN.

For each of ten seeds and both rewarded patterns, collect exactly 20,000 environment transitions using uniform random legal actions on the default maze. Reset at terminal outcomes or the 120-step limit. This shared behavior dataset prevents one controller from receiving a more favorable learning history.

Each controller receives the same 2,000 replay batches of 64 transitions. Tabular Q uses alpha 0.25 and gamma 0.95. Neural Q uses Huber TD loss, Adam learning rate 0.001, gamma 0.95, a frozen target copy refreshed every 100 updates, gradient norm clipping at 5, and nonnegative internal weights clipped to 3. No test-result tuning is allowed within this protocol.

All neural controllers share the input projection and initial readout for a given seed. No pretraining or privileged labels are used. The full graph is immutable; learned reduced-circuit weights go into separate checkpoints. Equal experience and update counts do not imply equal wall time or parameter count; both are reported.

Evaluation and decision rules

Part A: ten training seeds, rewarded A and B, 400 training episodes per condition, 100 frozen trials on the familiar maze and 100 on unseen layouts. Compare decoded tokens and exact Q-table hashes, and save matched outcomes. The current encoder resets for each of its two 40 ms probes, and does not evolve during movement.

Part B: evaluate before training and after 500, 1,000 and 2,000 updates. At each checkpoint run 100 trials on the familiar maze and 100 on unseen layouts, using a separate RNG seed namespace. Pair assignments, mazes and action-tie seeds across agents. Reusing frozen test conditions supports learning curves, not hyperparameter selection.

Exploration and learning are off during evaluation. Preserve policy hashes before/after, and fail if they change. Use 50 fixed unseen layouts with both pattern assignments. Exact x/y and adjacent walls remain observable, so this tests limited layout transfer rather than natural fly navigation or full-map planning.

Primary endpoint: familiar-maze success after the fixed budget. Report unseen-layout success separately, along with mean return, steps, successful-route efficiency, training time and trainable parameter counts. Compute paired differences averaged over both rewarded patterns within each training seed, with a bootstrap 95% interval over the ten seed units.

Call a controller better only on the named metric and tested setting; an interval overlapping zero is inconclusive. An anatomical benefit requires improvement over the matched randomized circuit as well as relevant tabular results. A failed neural learner, a tie or poor transfer must remain visible. This small benchmark cannot establish superiority over all ML/RL models.

Reproduction and scope updates

The comparison runner will save a prospective config, source manifest/archive, verified graph identity, selected circuit IDs/masks/weights, raw shared transition data, replay seeds, checkpoints, learning curves, trial outcomes, timing and a machine-readable report. Records remain separate from the live maze experiment.

Full-graph visual probes run with the requested existing CPU/CUDA backend. The small reduced-circuit benchmark uses NumPy on the GX10 host CPU; it does not silently claim GPU training. Dependency pins and source archives identify the implementation. Development smoke runs are excluded from the reported benchmark.

The protocol and its fingerprint are saved before collecting new outcomes. Any deviations or later changes to training budgets must be dated and reported, not retroactively described as prospective. Original Exp 1 dynamics and the current interactive Exp 2 learning rule are preserved.

Current setup defaults have changed since the original pages were written: `setup.sh`, `setup_exp_2.sh` and `setup_explorer.sh` now select real MaleCNS data. Synthetic software fixtures require `--dataset demo` explicitly. This setup correction does not change the original experimental results.

[01] Watkins and Dayan. Q-learning. Machine Learning (1992).

<https://www.gatsby.ucl.ac.uk/~dayan/papers/cjch.pdf>

[03] Lappalainen et al. Connectome-constrained networks predict neural activity across the fly visual system. Nature (2024).

<https://www.nature.com/articles/s41586-024-07939-3>

[04] MaleCNS official download inventory.

<https://male-cns.janelia.org/download/>

[05] Learning with reinforcement prediction errors in a model of the Drosophila mushroom body (2021).

<https://www.nature.com/articles/s41467-021-22592-4>

No comparative result from the new plastic network exists at the time of this amendment. Observed outcomes will be added to STATUS.md and the separate run report after verification.

Observed results: 21 September

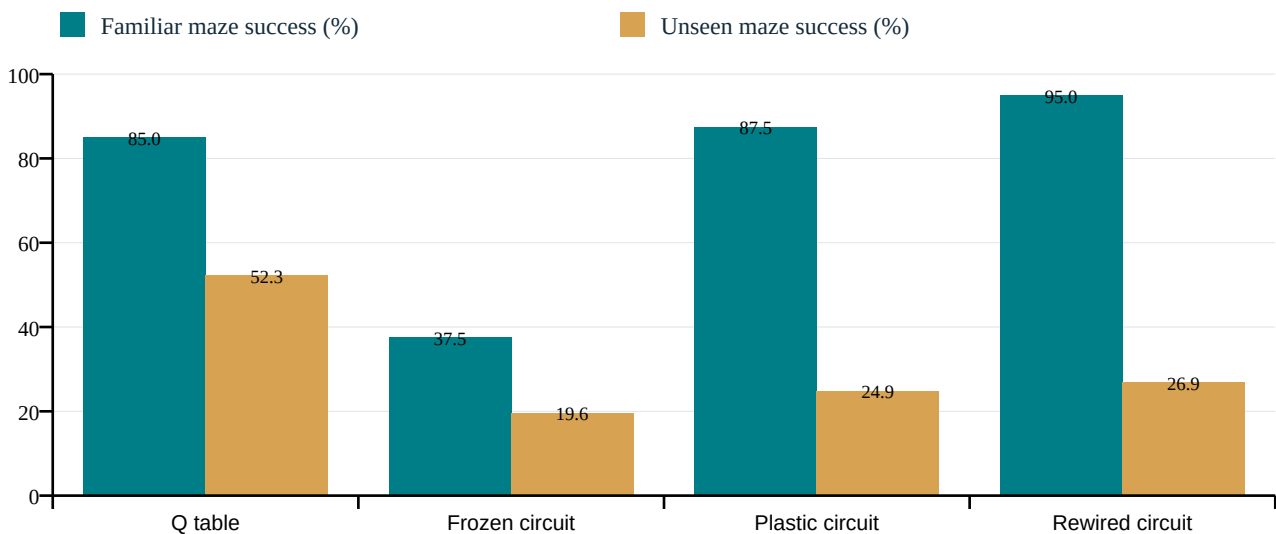
Written after the two full runs completed on 21 September 2026. This section reports observations; the preceding v0.2 amendment was authored before the new benchmark was implemented and run. No test-driven parameter tuning was performed.

Part A: the existing maze controller

Across ten seeds, both reward assignments and 400 online training episodes, fly-probe Q and direct-image Q produced exactly equal Q tables after every episode. Both achieved 2,000/2,000 familiar successes (100%) and 570/2,000 unseen successes (28.5%). Their difference was exactly zero. The neural probe supplies the same decoded tokens and adds no demonstrated value here.

Part B: equal recorded experience

Final success percentages after 2,000 replay updates. Each bar represents 2,000 frozen trials across ten training seeds and both reward assignments. These are point estimates; paired uncertainty is reported on the next page.



Part B uses shared random experience and a wall-aware state representation. Part A uses online training and the original state representation. Compare agents within each part; the different Q-table scores are not a contradiction.

What did training establish?

Connection strengths really changed

The selected anatomical circuit contains 256 Kenyon cells, 97 MBONs and 3,857 recorded positive KC-to-MBON connections. Training changed 3,013 to 3,283 permitted internal strengths per seed/reward condition, as well as the external action readout. Missing edges remained zero and all strengths remained nonnegative. The frozen-circuit control retained its internal weights exactly.

Familiar maze: no clear winner over Q-learning

The plastic circuit scored 87.5% versus 85.0% for the replay Q table: +2.5 percentage points, with a paired seed-bootstrap 95% interval of -12.5 to +17.5 points. That does not establish an advantage. It improved over the frozen-circuit readout at 37.5%, showing that training internal modeled strengths can matter in this particular reduced architecture.

Unseen layouts: the Q table transferred better

The replay Q table scored 52.3%; the plastic circuit scored 24.9%. Plastic minus Q was -27.4 percentage points, with a paired 95% interval of -35.7 to -18.05 points. Random movement scored 29.05%. The trained anatomical model did not demonstrate useful layout transfer in this experiment.

No demonstrated benefit from recorded wiring

The degree-matched rewired network scored 95.0% familiar and 26.9% unseen. Plastic minus rewired was -7.5 points familiar (95% interval -15.0 to 0.0) and -2.0 points unseen (-7.5 to +3.6). The recorded wiring did not outperform this matched control.

Conclusion: current Exp 2 is a tie between two equivalent Q learners. The new reduced plastic circuit learns, but these runs do not show that fly wiring is better than conventional learning. They also do not show that every possible fly-derived model must fail.

Limits, evidence and reproduction

The new model is a reduced feedforward rate network, not the full MaleCNS LIF simulation. Its input projection, tanh dynamics, action readout and gradient/Adam plasticity are engineered. There is no validated dopamine model, natural motor mapping or biological learning claim.

Ten optimization/experience seeds were tested, but only one sampled anatomical subcircuit and one degree-preserving rewiring. Intervals resample seed means after averaging rewarded A/B within each seed; they are not confidence intervals over all possible circuits. Hyperparameters and budgets were fixed rather than optimized separately for every model.

The existing 120-action timeout is treated as terminal, and remaining time is not an observation. This finite-horizon limitation applies to all compared agents. Test layouts are unseen, but absolute position and adjacent walls are observable. No full-map or shortest-path oracle enters a policy.

Approximate mean update-call times per condition on the GX10 host CPU: Q table 0.21 s, frozen-circuit readout 0.59 s, plastic circuit 0.70 s, rewired circuit 0.71 s. These exclude data collection, feature initialization and evaluation. Original full-graph probes used CUDA. Evaluation and immutable checkpoint checks passed.

Reproduce with the README comparison command. The compact, versioned report is docs/research/maze_comparison_results.md; its JSON supplies all learning-curve points and paired intervals. Raw experience, trials, circuit arrays and checkpoints remain outside Git under runs/gx10-a/exp_2_comparison and runs/gx10-b/exp_2_comparison.

Accepted run IDs: 20260921T060909Z-comparison-b15807 (original-controller audit) and 20260921T060903Z-comparison-a72cae (reduced-circuit comparison). Exact scientific source identity is recorded in each manifest and the compact report; immutable releases and a source archive preserve it.

Bookkeeping correction: the first runner hashed its mutable progress pointer before its final update. Explicit finalization_repair.json records retain the original manifests. All 378 immutable artifacts verified after excluding that pointer; no scientific output was changed.

Full-scale amendment: scope

21 September 2026, 06:32 UTC. Written before implementing or observing this full-graph learning test. The earlier reduced-circuit results are already known and remain above. This new test is separate from the original live maze service and leaves its behavior unchanged.

Use every node and CSR edge slot in the retained MaleCNS-derived graph: 166,700 neurons and 25,582,938 directed connections. Zero signed-weight entries remain zero. This is the entire prepared graph, not a claim to include every cell, synapse, physiology or body system of the living animal.

The full graph advances twice per maze observation, carrying neural state between actions and resetting at each trial boundary. A bounded recurrent rate model replaces LIF dynamics for this isolated comparison. Activity represents signed normalized deviations, not measured voltages, spikes or validated biological firing rates.

Inputs and outputs are engineered

The same 24 features are provided to all controllers: one-hot x and y, the assignment of two opposite pattern identities, and four adjacent legal-move bits. No reward identity, full maze, distance or path oracle is supplied. A fixed random signed channel assignment drives annotated sensory populations. Four learned linear action readouts pool all modeled neurons. These are explicit modeling choices, not natural fly sensory or motor pathways.

Full scale describes the retained network size and continuous use. It does not establish a complete functional reconstruction of a fly brain or biological learning.

Full network: dynamics and learning

Initialize each nonempty row of signed W to absolute sum 0.45. For two integration ticks, $h_{\text{next}} = 0.5 h + 0.5 \tanh(W h + \text{input})$. Each trainable weight retains its original sign and is clipped between zero and twice its initial magnitude. Thus the maximum absolute incoming sum stays at most 0.9, and activity remains bounded.

Train action values by semi-gradient temporal-difference learning: $\text{target} = \text{reward} + 0.95 \times \text{the best legal next-action value}$. Genuine food outcomes have no bootstrap; the 120-action time limit is treated as a truncation. Preview the next observation without committing neural state. Clip TD error to $[-1, 1]$.

The full plastic model updates every eligible internal edge using the derivative through both integration ticks of the current observation, with learning rate 0.01. Detach neural history before this observation. The action readout uses normalized updates: learning rate 0.25 divided by $1 + \text{squared activity norm}$. Bias receives the same normalization. No optimizer state, target network or natural dopamine rule is implied.

Frozen control: same graph and continuous state, but only the action readout trains. Plastic wiring control: shuffle presynaptic indices within positive and negative edge strata, preserving each row weight multiset and source degree counts within sign. This configuration-model multigraph may contain parallel edges and self-loops; report them and do not claim a simple-graph null.

Input projection and initial action readout are matched across the three neural conditions. Neural state is recurrent, but temporal gradients stop at each observation boundary. This is a bounded engineering model, not fitted fly physiology.

Full-scale test: fixed budget

Before outcomes: five seeds (0-4), both rewarded patterns, 10,000 uniformly random legal-action transitions per seed/reward condition, one chronological pass per learner. Preserve episode reset flags. All four trained conditions receive exactly the same experience: tabular Q, frozen full graph, plastic full graph and plastic rewired full graph. Add a random-action evaluation reference.

For each condition, evaluate the frozen final policy on 50 familiar-maze trials and 50 unseen-maze trials. The 25 unseen layouts each appear with both cue assignments. Test namespaces are disjoint from training. Neural evaluation advances the graph for every action; no observation-response cache is permitted. Manual actions never enter these records.

Primary endpoint is final familiar success; unseen-layout transfer is separate. Report returns, steps, route efficiency, timing, parameter counts, changed-edge counts, activity coverage and paired differences. Average both reward assignments within each seed, then bootstrap five seed units 10,000 times for descriptive 95% intervals. Five seeds give limited uncertainty resolution.

Before running behavior: verify numerical gradients on small graphs, CPU/CUDA agreement, reset/preview state isolation, sign bounds, frozen controls and finite whole-graph activity. Engineering timing may be measured without inspecting maze performance. No tuning or changing budgets after behavioral results are observed; failures and deviations are retained.

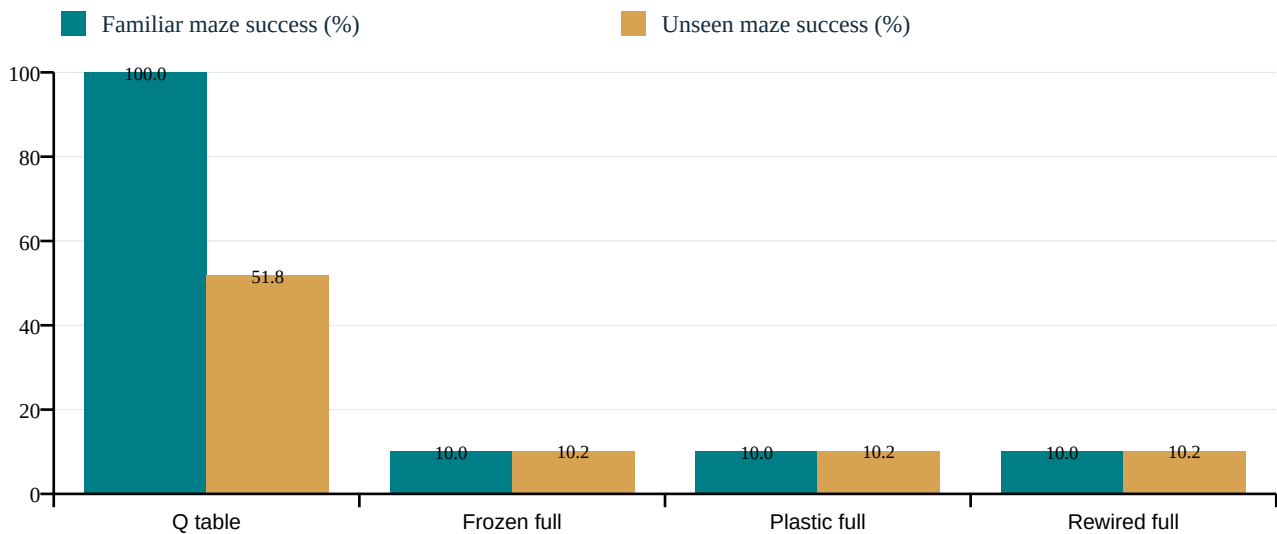
Save the exact protocol/PDF fingerprint, source archive, graph hashes, input mapping, rewired indices, raw experience, all final policies, trial records and immutable artifact hashes. Runs are isolated on both GX10s; they do not distribute one brain across machines. Compare within this protocol, not directly with the earlier different-budget benchmark.

No outcome exists for this full-scale protocol when these pages are authored. A negative result remains useful: it tests this stated model and training budget, not every possible full-brain model.

Observed full-scale results

Written 21 September 2026, 07:20 UTC, after all primary runs and the separate diagnostic finished. The v0.4 full-scale protocol was saved and committed before behavioral runs began. The original reduced-circuit and cached-probe results above remain separate experiments.

The fixed budget covered five seeds, both reward choices and 10,000 identical recorded training transitions per condition. Each of five agents had 500 familiar and 500 unseen frozen trials in total. The neural conditions advanced the full graph on every action, with no pattern-response cache.



Tabular Q: 100.0% (500/500) familiar and 51.8% (259/500) unseen. Full plastic graph: 10.0% (50/500) familiar and 10.2% (51/500) unseen.

Random movement reference: 31.2% (156/500) familiar and 28.4% (142/500) unseen.

The full-connectome model did not beat tabular Q under this stated budget. This compares an engineered full-graph model with a Q table; it is not a test of a living fly or every possible connectome-based learning method.

How full was the simulation?

Every neural decision advanced all 166,700 retained neurons and all 25,582,938 CSR edge slots. Neural activity persisted between actions and reset at trial boundaries. The entire prepared graph was used, including quiet cells. This is the retained dataset, not a claim to reconstruct every physiological detail of an animal.

Of the edge slots, 24,469,412 began with nonzero signed strength. The other slots stayed zero. Engineered position, cue-assignment and wall features drove 17,885 annotated sensory neurons. Four learned action-value readouts pooled the modeled population.

During anatomical-model training, 142,850 to 143,092 neurons exceeded the reporting threshold of 0.000001 normalized activity. A descriptive graph audit found 166,614 neurons structurally reachable from the input population and 86 unreachable. Reachability does not guarantee strong activity: attenuation, cancellation and unused input channels can keep cells quiet.

Strengths changed, but only slightly

Training changed 8,383,190 to 8,688,335 existing internal strengths in each full-plastic condition. However, the norm of the total weight change was only 0.001691% to 0.002515% of the initial weight norm. A large count of changed entries should not be mistaken for a large functional adaptation.

Frozen internal weights remained exactly unchanged; their action readouts trained. All plastic variants retained the declared signs and bounds. These checks establish implementation behavior, not biological learning.

This model uses bounded signed rate deviations, two integration ticks per observation and a short semi-gradient TD rule. It does not simulate validated spikes, chemical signaling, natural dopamine plasticity or fitted fly physiology.

Paired comparisons and limits

Full plastic graph versus Q table

Familiar success difference: -90.0 percentage points (paired 95% interval -100.0 to -80.0). Unseen success difference: -41.6 percentage points (paired 95% interval -46.4 to -35.8).

Does plasticity or recorded wiring help?

Plastic minus frozen graph: +0.0 percentage points (paired 95% interval +0.0 to +0.0) familiar; +0.0 percentage points (paired 95% interval +0.0 to +0.0) unseen.

Anatomical plastic minus rewired plastic: +0.0 percentage points (paired 95% interval +0.0 to +0.0) familiar; +0.0 percentage points (paired 95% interval +0.0 to +0.0) unseen.

Intervals resample five seed means, after averaging both reward assignments within each seed. Familiar trials repeat one layout and two cue assignments; 500 trials are not 500 independent training replications. Only one input mapping and one sign-stratified wiring null were used.

The null preserved each row weight multiset and sign-stratified source degree counts, but allowed parallel pairs and self-edges. It contained 74,530 excess parallel edges and 297 self-edges. Unseen mazes changed walls but retained start and food coordinates. No path or reward identity entered the policy.

No test-result tuning was performed. The single-pass budget, very small internal changes, directly accessible sensory readout and short gradient history limit what this result says about stronger learning methods. Network size alone does not establish useful task computation.

Does the policy need recurrence?

This diagnostic was defined after partial primary outcomes revealed matching neural success rates. It is explicitly post hoc. The primary models, seeds, experience, hyperparameters and evaluation were left unchanged.

For every saved full-plastic policy, remove all recurrent strengths and retain its input mapping, learned action readout and bias. Reset at each trial, retain the same two-tick rate equations, and repeat all 1,000 primary trial definitions. No retraining occurs. The zero sparse product is omitted algebraically for this diagnostic only.

Complete paths matched on 989/1000 trials, and terminal outcomes matched on 995/1000. These are direct paired comparisons with the saved full-plastic controller, checked again against raw trial records.

The diagnostic ran in NumPy float32 on the GX10 host CPUs. Its zero-recurrence equations were checked against the sparse CPU reference. It must not be confused with the primary CUDA experiment, which actually computed through the full graph.

Most paths remained identical despite removing recurrence, supporting sensory/readout dominance in this setup. The small differing subset also changes numerical backend (CPU versus CUDA); rounding may contribute, so it is not a pure causal estimate of recurrence. Unchanged success alone does not prove that a circuit has no effect.

A stronger follow-up would require a new dated protocol, training/validation separation, stronger verified internal adaptation and a readout that cannot directly bypass circuit processing. Such a follow-up was not silently folded into this comparison.

Full-scale evidence and reproduction

Independent verification checked 64 immutable primary artifacts, all 30 full neural checkpoints and all 5000 saved primary maze trajectories. Saved weight arrays matched recorded hashes, signs and bounds. Checkpoint reload reproduced the first familiar trial in each neural condition, and frozen evaluations preserved parameter hashes.

Mean training time per condition, excluding initialization/evaluation: q_table: 0.02 seconds, full_fixed: 42.69 seconds, full_plastic: 114.26 seconds, rewired_plastic: 217.28 seconds. Equal experience did not mean equal compute.

Trainable parameter counts: Q table 10,368; full frozen-graph readout 666,804; full plastic model 25,136,216. The two GX10s ran independent reward-condition partitions, not one distributed brain.

Accepted primary runs: 20260921T063920Z-fullscale-e4d345; 20260921T063951Z-fullscale-29e490. Exact scientific source SHA-256: 1dd798bb860a282714893a48033d73346b47c9789ec8188ffe508e74217357e6. The pre-outcome implementation/protocol commit is 9fccab1.

README.md documents build, run, partition and verification commands. The compact machine-readable report is docs/research/fullscale_results.json, with a readable .md companion. Raw experience, neural policies, source archives, mappings and trial paths remain outside Git under runs/.

The post-hoc diagnostic has its own config, source archive, raw trial files and recorded source identity. The report distinguishes it from the prospective experiment. CPU/CUDA agreement, gradient checks and the existing regression suite passed. Original Exp 1, live Exp 2 and the anatomical viewer remain unchanged.

Keep one current exp_2.pdf in PLAN. Earlier protocol versions remain recoverable from Git, and immutable run/source archives preserve the code actually used for the science.